A Small Experiment on LongMemEval-S

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1 Introduction

This experiment compares three conditions: no history, recent history, and BM25 retrieval. Each condition provides different history to the model, which then answers the same questions. I count the correct answers and analyze why the model fails. The dataset is the cleaned LongMemEval-S dataset.[1] This note shares a basic understanding of the code, questions, and experimental results.

2 About the Code

I set the history token budget to 8,000 for both recent history and BM25. This limits the amount of history selected for each question. It does not include the question, answering instructions and the reference. The actual history length should be below this limit.

The three conditions work as follows:

1. **No history:** The model receives the question and instructions, but no chat history. In the code, the selected session list is empty.

2. **Recent history:** The code checks sessions from newest to oldest and selects whole sessions that fit within the remaining budget. This condition uses recency, not relevance to the question.

3. **BM25:** The code scores sessions by matching words in the question with words in each session. The advantage is an exact score for each session, making them easy to sort. The score considers word frequency, how common a word is across sessions, and session length. Repeating the same word gives smaller additional gains.

   For both recent history and BM25, if a whole session does not fit within the remaining budget, the code skips it and checks the next one.Selected sessions are then presented in chronological order.

3 Questions and Results

Five questions were randomly selected from each of five question types, giving 25 questions in total.

I first used `gpt-6-luna`. I then ran an additional experiment with `gpt-6.1-sol`, using medium reasoning effort and the same 25 questions(75 requests) and selected histories. Accuracy has obviously improved. Hypothesizing that reasoning effort may have a positive impact on the accuracy, then I added an experiment with `gpt-6-luna`, using medium reasoning, only testing the performance only with BM25. Table 1 shows the results for different question types. Table 2 shows the accuracy of different models, mean hisotry tokens, and mean output tokens including both reasoning tokens and final answer tokens. Interestingly, Sol consumes fewer tokens than Luna with medium reasoning while achieving the same accuracy. Answers were checked manually against the reference answers,allowing equivalent wording.

[1] `https://huggingface.co/datasets/xiaowu0162/longmemeval-cleaned`

Model	Effort	History	SS-U	SS-A	KU	TR	MS
Luna	none	No history	0/5	0/5	0/5	0/5	0/5
Luna	none	Recent	0/5	1/5	0/5	0/5	0/5
Luna	none	BM25	5/5	5/5	3/5	0/5	0/5
Luna	medium	BM25	5/5	5/5	3/5	3/5	2/5
Sol	medium	No history	0/5	0/5	0/5	0/5	0/5
Sol	medium	Recent	0/5	1/5	0/5	0/5	0/5
Sol	medium	BM25	5/5	5/5	3/5	3/5	2/5

SS-U: single-session-user; SS-A: single-session-assistant; KU: knowledge-update; TR: temporal-reasoning; MS: multi-session.

Table 1: Correct answers by question type.

Model	Effort	History	Accuracy	Mean history tokens	Mean output tokens
Luna	none	No history	0/25 (0%)	0	/
Luna	none	Recent	1/25 (4%)	7,858.96	/
Luna	none	BM25	13/25 (52%)	7,891.80	18.28
Luna	medium	BM25	18/25 (72%)	7,891.80	67.12
Sol	medium	No history	0/25 (0%)	0	/
Sol	medium	Recent	1/25 (4%)	7,858.96	/
Sol	medium	BM25	18/25 (72%)	7,891.80	31.80

Table 2: Overall accuracy, mean history length, and mean output tokens.

BM25 performs better than recent history on this set. With the same selected history, Sol and Luna(medium) answer five additional BM25 questions correctly, increasing accuracy by 20 percentage points.

4 Analysis of Failures

I group the observed failures into three categories. Because of space limits, I picked out some representative mistakes.

4.1 Enough Evidence, but an Incorrect Answer

Sol and Luna(medium) corrected five Luna(none) failures using the same input. Two examples are shown below:

- `gpt4_2312f94c`: The question asks whether the phone or laptop was obtained first. The phone was obtained on February 20, while the laptop actually arrived on February 25. Luna(none) answered incorrectly.
- `e25c3b8d`: The original handbag price was $500, and the user paid $200. The saving was therefore $300. Luna(none) treated the prices as conflicting information instead of calculating the difference.

These examples show that retrieving useful evidence is not always enough. The model also needs to interpret dates, distinguish different kinds of information, and combine facts correctly.

4.2 Only Part of the Evidence Was Selected

For `5a7937c8`, the input includes faith-related activities on December 10 and 17, but misses the Christmas Eve mass on December 24. The missing session ranked fourth and needed 2,546 tokens, while only 1,113 tokens remained. Sol answered two days rather than three.

4.3 Key Evidence Was Not Selected

For `gpt4_8279ba03`, the question asks which kitchen appliance the user bought ten days earlier. The question date is March 25. A March 15 session says the user got a smoker that day, but this session ranked 22nd and was not selected.The retrieval code does not convert "ten days ago" into a target date. It also relies on word matching, so phrases such as "kitchen appliance" and "buy" may not match "smoker" and "got" well.

4.4 Ambiguous Evidence or Reference Answers(additional)

For `a2f3aa27`, the history contains a confirmed Instagram follower count of 1,250 and a later statement that the user thinks they are close to 1,300. The reference answer is 1,300. Sol distinguished the confirmed count from the later estimate. I counted it as incorrect under the reference answer, but this is not a clear reasoning failure.

5 Conclusion

The failures suggest that selecting useful information is a core challenge. The results for Luna (medium) and Luna (none) also show that reasoning effort is an important factor in question answering.

The results suggest two main limits: whether the selected history contains enough evidence, and whether the model can use that evidence correctly. Sol corrected five cases with sufficient evidence, but several remaining failures involved missing history. Possible next steps include improving retrieval by considering semantic meaning and connections between sentences, rather than relying only on word matching. Another direction is to organize the history more efficiently.